	classical	prob.	quantum
state	n bits	2^n probs. / n bits	2^n amplitudes / n qubits
operations	implementable by TM	plus prob.	reversible, unitary
output	n bits	sample from P_x	sample from α_x^2
implementation	transistors	simulation physical	superconducting circuits, ions, photons, spin of electrons, topological qubits

Hadamard

$|0\rangle \rightarrow \sqrt{1/2}$ ampl. on $|0\rangle$
$\sqrt{1/2}$ ampl on $|1\rangle$

$|1\rangle \rightarrow \sqrt{1/2}$ ampl. on $|0\rangle$
$-\sqrt{1/2}$ ampl. on $|1\rangle$

Q Coin Flip

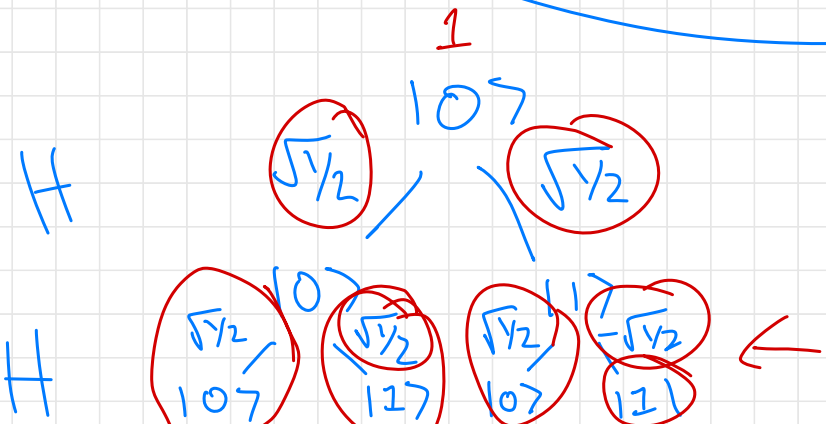
new qubit A — 1 ampl. on $|0\rangle$, 0 amp. on $|1\rangle$

Hadamard A — $\sqrt{1/2}$ ampl. on $|0\rangle$, $\sqrt{1/2}$ ampl. on $|1\rangle$

Hadamard A — $1/2$ ampl on $|0\rangle$, $1/2$ ampl on $|1\rangle$, $1/2$ ampl on $|0\rangle$, $-1/2$ ampl on $|1\rangle$

$\Rightarrow$ 1 ampl on $|0\rangle$, 0 ampl on $|1\rangle$

extract all — ~~0 w/ prob $\sqrt{1/2}^2 = 1/2$~~, ~~1 w/ prob $\sqrt{1/2}^2 = 1/2$~~



$\sqrt{1/2} \cdot \sqrt{1/2}$ $\quad$ $\sqrt{1/2} \cdot \sqrt{1/2}$ $\quad$ $1/2$ $\quad$ $-1/2$

$= 1/2$ $\quad$ $= 1/2$